EET202

HINES & TRANSFORMERS

MODULE:2

4 – Armature Reaction (contd..)

e performance characteristics of DC generators

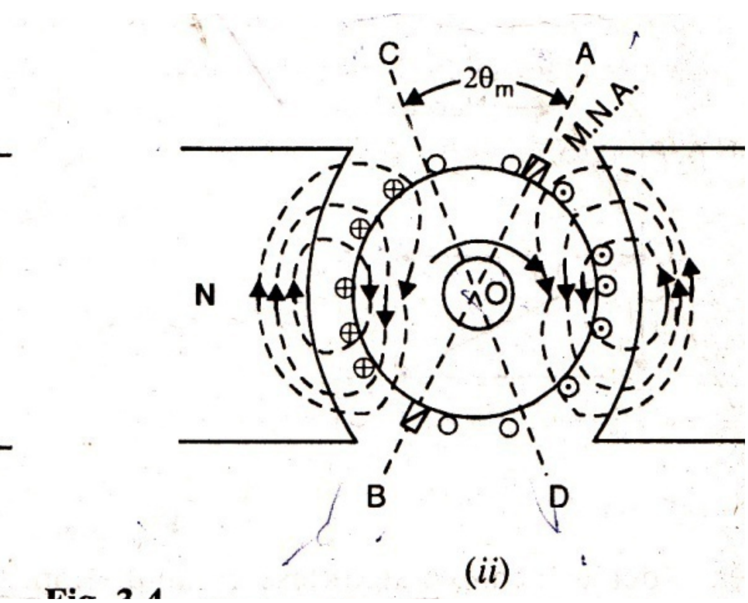
Prepared By,

ADARSH SR
Asst. Professor
EEE Department
SCET Kodakara

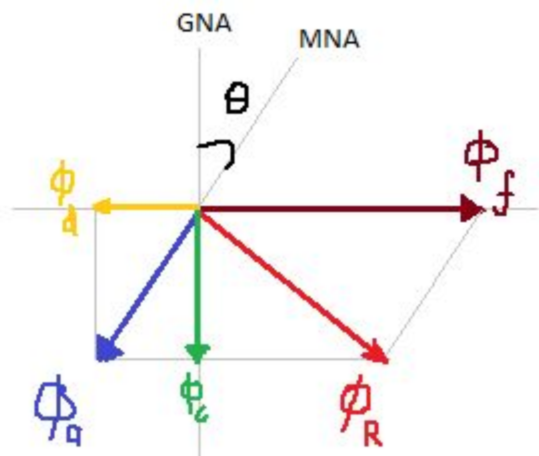
sing & Cross-magnetising Effect

conductors ϴ degree on either side of GNA n direct opposition to main flux. Thus the g within angles AOC=BOD=2ϴ at the top and ture produce demagnetising effect and these are **tising armature conductors**.

netisation of armature conductors lying between l COB is at right angle to the main flux. These duce cross-magnetising effect and are called **ng armature conductors**.

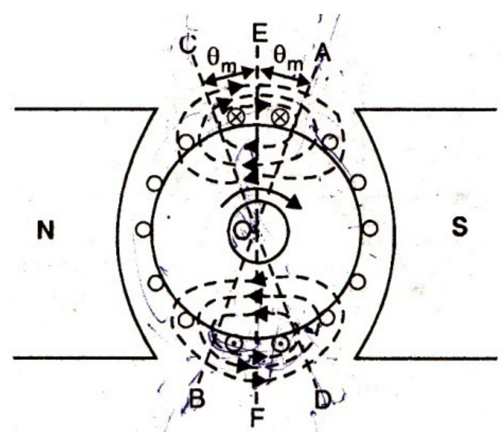


(ii)



demagnetising Ampere-Turns/Pole

gnetising Ampere-Turns(AT) can be neutralized by
he main field windings.



ber of armature conductors

ch armature conductor

wave winding

lap winding

magnetising conductors = $\frac{4\theta_m}{360} * Z$

g AT = $\frac{1}{2}\left[\frac{4\theta_m}{360} * Z\right] * I$ $\frac{2\theta_m}{360} * ZI$

a pair of poles.

mpere-Turns/pole, AT$_d$/pole = $\boxed{\frac{\theta_m}{360} * ZI}$

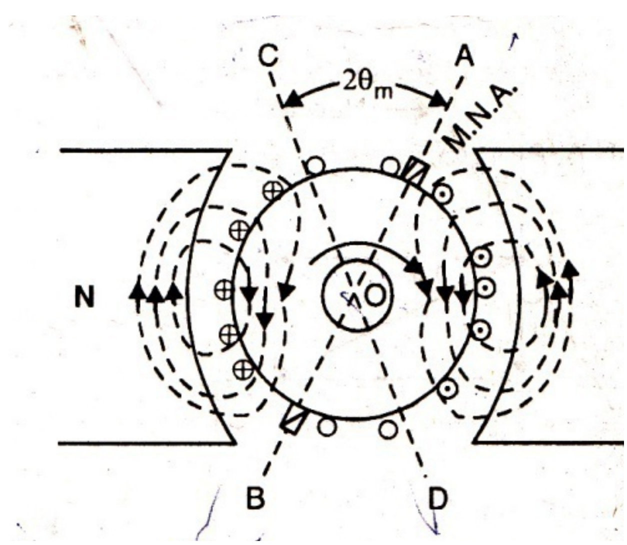
ns to be added in main field winding= $\frac{AT_d / pole}{FieldCurrent}$

ross-magnetising Ampere-Turns/Pole

= = $\frac{Z/2}{P} * I$ $\frac{ZI}{2P}$

g AT = $\frac{\theta_m}{360} * ZI$

ng Ampere-Turns/pole, AT_c/pole = $\boxed{\left[\frac{1}{2P} - \frac{\theta_m}{360}\right] ZI}$



Z = total number of armature conductors

I = current in each armature conductor

$= I_a/2$ ... for simplex wave winding

$= I_a/P$ ... for simplex lap winding

θ_m = forward lead in mechanical or geometrical or angular degrees.

ure conductors in angles AOC and BOD is $\frac{4\theta_m}{360} \times Z$

nstitute one turn,

er of turns in these angles $= \frac{2\theta_m}{360} \times ZI$

mp-turns per pair of poles $= \frac{2\theta_m}{360} \times ZI$

gnetising amp - turns/pole $= \frac{\theta_m}{360} \times ZI$ $\quad \therefore AT_d$ per pole $= ZI \times \frac{\theta_m}{360}$

tising AT per pole

between angles AOD and BOC constitute what are known as distorting or ors. Their number is found as under :

ctors/pole both cross and demagnetising $= Z/P$

gnetising *conductors*/pole $= Z\frac{2\theta_m}{360}$ (found above)

gnetising *conductors*/pole $= \frac{Z}{P} - Z \times \frac{2\theta_m}{360} = Z\left(\frac{1}{P} - \frac{2\theta_m}{360}\right)$

sing *amp-conductors*/pole $= ZI\left(\frac{1}{P} - \frac{2\theta_m}{360}\right)$

agnetising *amp-turns*/pole $= ZI\left(\frac{1}{2P} - \frac{\theta_m}{360}\right)$

(Remembering that two conductors make one turn)

$$AT_c/\text{pole} = ZI\left(\frac{1}{2P} - \frac{\theta_m}{360}\right)$$

the demagnetising effect of armature-reaction, an extra number of turns may be

No. of extra turns/pole $= \frac{AT_d}{I_{sh}}$ —for shunt generator

npensating Windings

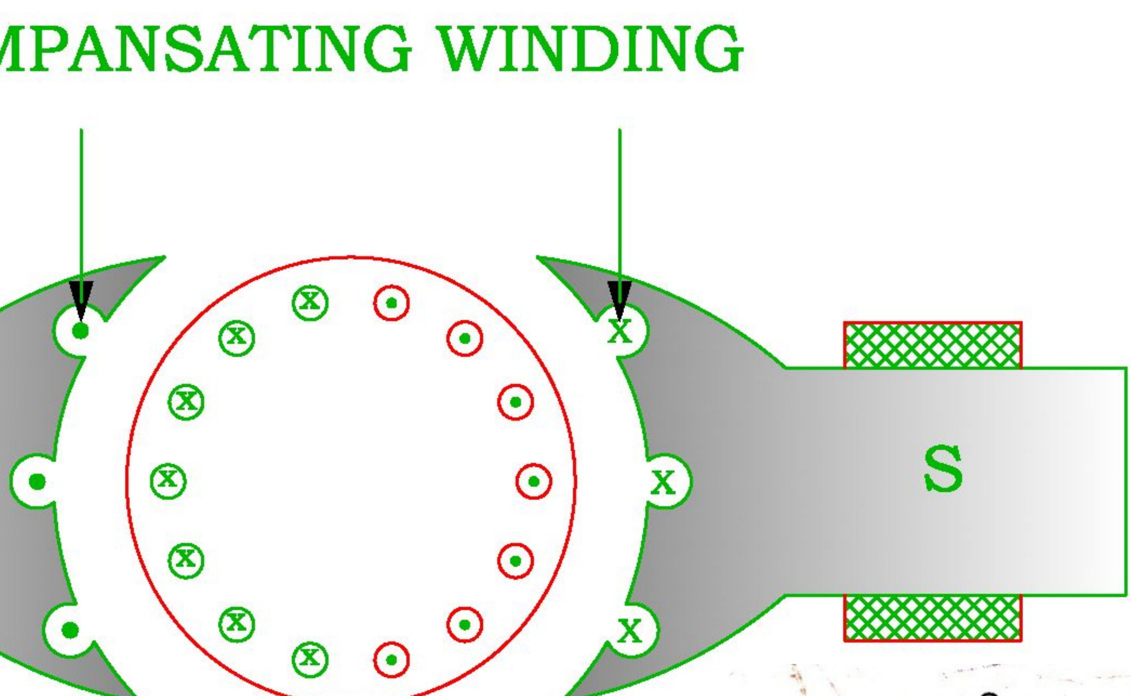
tising effect leads to sparking at the brushes.

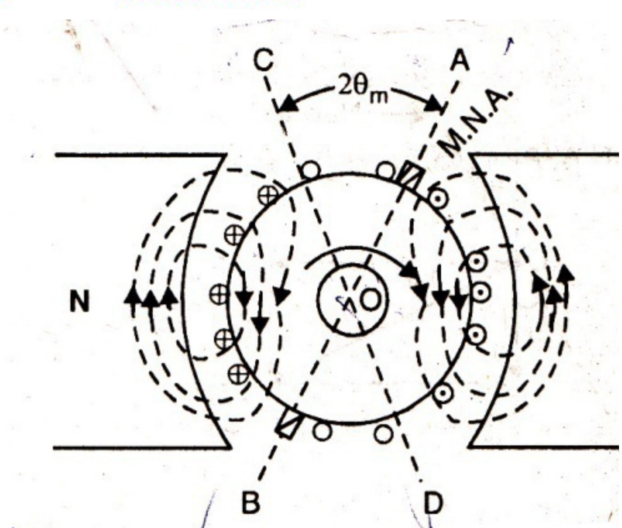
utralise cross-magnetising effect of armature sating windings are used.

indings are the auxiliary windings embedded in faces.

ted in series with armature in a manner so that current through compensating conductors in any l be opposite to the current through the adjacent ctors.

npensating Windings





s/Pole for Compensating Winding

ure conductors /pole = Z / P

ure turns /pole = Z / 2P

ure turns under one pole face = $\frac{Z}{2P} * \frac{PoleArc}{PolePitch}$

ole required for compensating winding =

$\frac{PoleArc}{PolePitch}$ = Armature AT/pole * $\frac{PoleArc}{PolePitch}$